

Enhanced CatBoost with Stacking Features for Social Media Prediction

Shijian Mao
maoshijian@corp.netease.com
UX Center, NetEase Games
Shanghai, China

Gaotian Lü
lvgaotian@corp.netease.com
UX Center, NetEase Games
Shanghai, China

Wudong Xi
xiwudong@corp.netease.com
UX Center, NetEase Games
Guangzhou, China

Xingxing Xing
xingxingxing@corp.netease.com
UX Center, NetEase Games
Guangzhou, China

Lei Yu
yulei01@corp.netease.com
UX Center, NetEase Games
Guangzhou, China

Xingchen Zhou
zxcn4056@corp.netease.com
UX Center, NetEase Games
Hangzhou, China

Wei Wan*
gzwanwei@corp.netease.com
UX Center, NetEase Games
Hangzhou, China

ABSTRACT

The Social Media Prediction (SMP) challenge aims to predict the future popularity of online posts by leveraging social media data. Social media data contains multimodal information, such as text, images, time series, etc. Previous methods have proposed many feature extraction and feature construction methods to represent these multimodal information, thereby predicting the popularity of posts. Despite the success of previous methods in extracting features from social media data, these features tend to be predominantly lower-order, posing a challenge in accurately capturing the rich information contained in text and images. In this paper, we propose a more diverse feature mining method and introduce a stacking block module to capture higher-order feature information contained in text and images. "lower-order" refers to the original high-dimensional embedding representation, while "high-order" pertains to the impact on post social popularity captured by tree models from text or image. We conducted massive experiments to evaluate the effectiveness of our proposed method and found that the stacking block module significantly improved performance.

CCS CONCEPTS

• **Information systems** → **Multimedia and multimodal retrieval**; • **Computing methodologies** → *Computer vision tasks*.

KEYWORDS

Social Media, Stacking Features, Popularity Prediction

*Wei Wan is the corresponding author.

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MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

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ACM ISBN 979-8-4007-0108-5/23/10...\$15.00
<https://doi.org/10.1145/3581783.3612839>

ACM Reference Format:

Shijian Mao, Wudong Xi, Lei Yu, Gaotian Lü, Xingxing Xing, Xingchen Zhou, and Wei Wan. 2023. Enhanced CatBoost with Stacking Features for Social Media Prediction. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3581783.3612839>

1 INTRODUCTION

With the rapid development of the Internet, social media platforms such as Weibo, WeChat, Facebook, Instagram, etc. have become the main ways for people to share information in their daily lives [4, 17, 32]. Massive social media data contains various information such as users' interests, opinions, behaviors, etc., and it has received extensive attention from industry and academia [11, 23]. A large amount of data from social media has been applied to various research tasks, leading to great breakthroughs in social media-related technologies and applications, such as Visual Genome [12], YFCC [10], Yelp2016 [1], etc.

The purpose of the social media prediction task is to predict the future popularity of new online posts via social media data. Social media data encompasses diverse types of information, including text, images and statistical features. To achieve precise social media predictions, a combination of natural language processing, image recognition and machine learning methods is needed [28, 29]. In terms of this task, previous methods [3, 13, 25, 31] have delved into feature extraction, feature intersection and model combination. However, the features they extract are predominantly lower-order, making it challenging to capture the wealth of information inherent in social media data images and text with just lower-order feature input models.

In addition to basic statistical features, we utilized cross-features to more accurately distinguish between post features and user preferences for both posts and users. For instance, popular topics of posts on the internet exhibit significant variation across different time periods. In general, the online sharing behavior of users is influenced by network environment. To capture these fluctuations, we computed the proportion of releases across various categories

within specific time intervals. To characterize users' sharing habits, we extracted relevant features from different categories of posts previously published by the users.

Compared to constructed feature information, the text and images in posts provide more specific content information and play a more significant role in predicting post popularity. Our proposed method places more emphasis on the text and images of the post. An instance is when tag information filled in by a user does not match image information due to unintentional or intentional amplification of platform exposure. To mitigate this issue, we employed a model to learn embedding representations of all tags and images, and subsequently identified tags with the highest image similarity. We propose an innovative approach to learn higher-order features of both text and images. Our method involves separately inputting text and images into natural language processing and image processing models to extract their respective embedding representations. We use cross-validation to obtain prediction results based solely on embeddings from the training and testing datasets and subsequently utilize these predictions as stacking features, which effectively represent the information of both text and images [9].

The main contributions of this work are as follows.

- We proposed an Enhanced Catboost with Stacking Features method for social media prediction, named "ECSF". To enhance the information representation of posts and users, we utilize various feature extraction and construction techniques.
- We novelly introduce a novel stacking feature to capture higher-order information from both text and images.
- Experimental results demonstrate the effectiveness of our rich feature extraction and stacking feature in considerably improving the accuracy of predicting post popularity.

2 RELATED WORK

Due to the abundant multimodal data and dynamic attributes, predicting the popularity of social media content has become a popular research topic, motivating many scholars to investigate feature extraction and model construction methods [2, 3, 13, 16, 25, 31]. [16] proposes a learning-to-rank method that is not only capable of addressing the sparsity in the training signal, but also scalable enough to enable real-time training. In [13], various types of features are artificially constructed based on social media data to capture their multimodal information. While these methods have exhibited some degree of success, they have not yet explored higher-order text and image representations, thereby limiting their ability to effectively capture the content and characteristics of social media posts.

As mentioned before, we proposed an enhanced Catboost with stacking features method for social media prediction. The effectiveness of stacking features depends on the advanced capabilities of multimodal representation learning, which enables accurate representation of complex data across multiple modalities. The progress in natural language processing [5, 18, 20, 24], image recognition [6, 7, 14] and other relevant technologies has spurred the continued advancement of multimodal representation learning [8, 15, 19], increasing its ability to effectively capture complex information of multimodal data. [22] presents a pre-trained neural network architecture for image-text matching, which has attained

state-of-the-art performance across multiple tasks in the field of multimodal research.

3 METHOD

To effectively capture user and post characteristics, we have developed various types of post profile features that encompass both user-level and post-level features. In order to comprehensively capture the textual and image information in a post, we have employed a stack block module to generate stacking features in addition to manual processing features. These features are then concatenated and fed into the CatBoost [21] model to derive the predicted popularity score for the post. The overall framework of proposed method is shown in Figure 1.

3.1 Stacking Block

In order to better capture the higher-order feature information of text and images, we innovatively propose a stacking block module. The implementation process of the stacking block is depicted in Figure 2: (1) Initially, some appropriate pre-training models are chosen for text or images to obtain their embedding representations; (2) Then, the embeddings are utilized as input features while the post popularity scores are used as the label to construct a dataset; (3) The dataset is then divided into training datasets and validation datasets using 5-fold cross-validation approach; (4) Finally, CatBoost model is utilized to train the training datasets. And the predicted scores outcomes from the validation set of each fold are used as stacking features, which will be concatenated with the original features.

For the text p_{text} of the post p , we obtain the stacking features $F_{p_{text}}$ as follows,

$$E_t = [M_1(p_{text}), \dots, M_k(p_{text})],$$

$$F_{p_{text}} = [F_{ct}^{fold1}(E_t^1), F_{ct}^{fold1}(E_t^2), \dots, F_{ct}^{fold5}(E_t^k)] \quad (1)$$

where M_k denotes the k -th pre-training model, E_t^k denotes the embedding representations obtained by the k -th pre-training model, and F_{ct}^{fold1} denotes the Catboost model for the *fold1* data.

3.2 Post Profile

To deeply analyze the similarities and differences between different posts, we construct various types of basic features for post profiles through artificial means. Similar to paper [13], we construct the basic time features of the post, such as the day of the week, the hour of the day, etc., and the basic features of the text, such as the number of words, the number of tokens, etc. Referring to paper [30], we globally sort some existing features and use the order as a ranking feature.

Furthermore, we have carefully considered numerous features that can indicate the nature of the post and its environment. First of all, we calculate the ratio of posts published in different categories within that period to determine the popularity of a category during this period. Additionally, we counted the total number of posts within a certain period to assess the website's level of activity during that time. To capture diverse information about posts published on the website during different time periods, we calculated the proportion of hot tags within each period. Considering that the popularity of posts changes over time, we extracted time-series

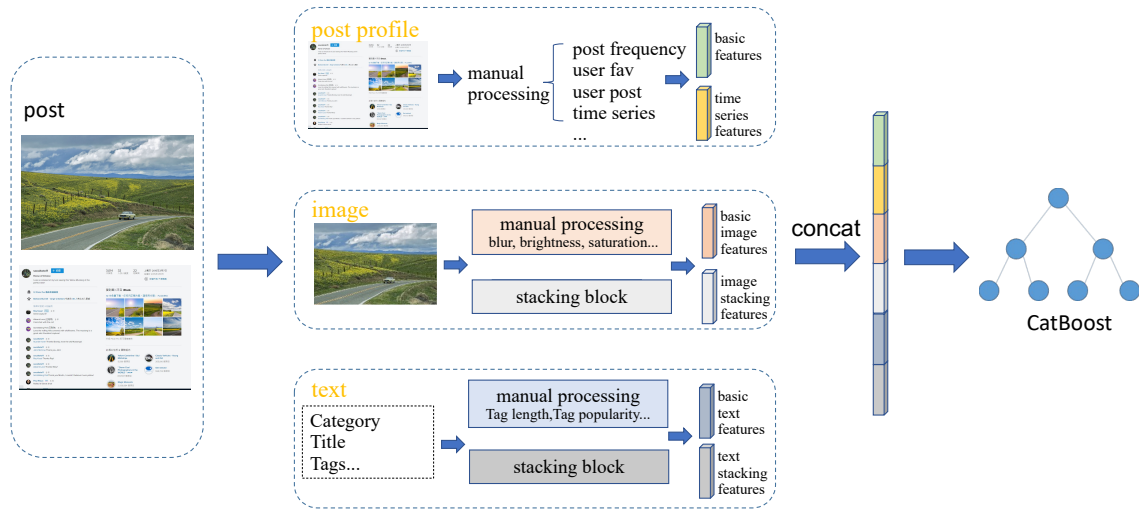


Figure 1: Illustration of the proposed method.

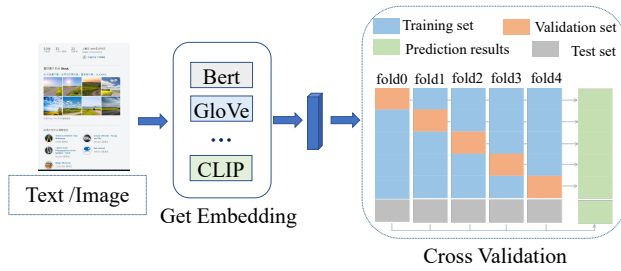


Figure 2: Illustration of the stacking block.

related features based on user structural features. We extract time-series features from the raw data by grouping them by user and ordering them based on time, including features such as the number of posts published in a given interval and the time elapsed since the last post was published.

We believe that the characteristics of the users who publish the post will also have a greater impact on the popularity of the post. On the basis of the original user information, we crawled three types of data as user supplementary features. The first type of data is the user's basic portrait data, including the number of fans, followers, and total number of photos posted [13]. These features provide a dimensional description of the user's overall influence. The second type of data is the user's historical posting photos data, including photo tag, title, and number of views, which enables us to capture the user's posting behavior, historical posting frequency, preferred posting types, and the heat of various types of photos posted. Lastly, the user's preferred photo data, including liked photos' tag, title, and number of views, is utilized to capture the user's preference.

3.3 Image Features

Image data can make people understand what the post wants to express at a glance, and attract the attention of other users. Regarding image data, we manually process and obtain basic image information such as brightness, blur, saturation and contrast. We utilize CLIP [22] to obtain the image and text embedding, and calculate the similarity between them as the auxiliary feature of the image. More importantly, we obtain image embeddings using the CLIP model, and subsequently use cross-validation to predict results on both the training and test sets based solely on the embeddings. We then utilize these prediction results as features and integrate them into the original input. Through the stacking block module of the image, we can effectively capture the high-order information of the image, and alleviate the limitations of tree models such as CatBoost in fitting high-dimensional embedding features.

3.4 Text Features

In the text information of the post, such as title, tag, etc., generally describe the specific content that the post wants to express. Basic text features, such as tag length, tag popularity, etc., are obtained through manual processing. We utilize the GloVe [20] model to obtain tag embeddings and reduce their dimensionality to 32 dimensions using Principal Component Analysis (PCA), allowing for diverse text representation. In the stacking block module for text, we utilize richer models such as Word2Vector [18], GloVe, Bert [5] and CLIP to generate text embeddings, and integrate their prediction results into the input to enhance the model's efficacy.

3.5 Regression Model

Gradient boosting is a highly effective machine learning technique that has achieved state-of-the-art results in a range of regression tasks. Since social media data contains more categorical features,

Table 1: The Overall results with divergence features.

Model	SRC	MAE
Official Basic Features	0.651	1.383
+User Supplemental Features	0.695	1.292
+Tag Embedding	0.730	1.233
+Time Series Features	0.743	1.175
+Text-Image Similarity Features	0.745	1.163
+Stacking Features	0.764	1.124
+User Interaction Features	0.768	1.094

CatBoost adopts a more effective strategy for learning category feature information than LightGBM [33]. In our proposed method, we concatenate all features and input them into CatBoost for predicting the post's popularity score.

4 EXPERIMENT

4.1 Dataset

The dataset was collected from a large photo sharing website Flickr for social media research, named Social Media Prediction Dataset (SMPD)¹ [26, 27]. SMPD contains over 69K online users, who share 486K social multimedia posts in Flickr. The aim of the dataset is to predict the temporal popularity of newly shared posts, where popularity refers to the future interactions or attraction of the post.

4.2 Evaluation Metrics

Following paper [26], to fairly compare the performance of the models, we employ two common metrics for validation, namely Spearman Ranking Correlation (SRC) and Mean Absolute Error (MAE). SRC is applied to measure the correlation between the predicted popularity ranking and the ground-truth popularity ranking. The aim of MAE is to measure the accuracy of the predicted scores, which is expressed as follows:

4.3 Overall Results

To evaluate the efficacy of our proposed multimodal features, we will employ CatBoost [21] algorithm to compute their prediction performance in the experiments. Follow [13], CatBoost with K-Fold cross validation can effectively improve model performance. The overall results are shown in Table 1.

First of all, we fed the official basic features [26] into a CatBoost model to predict the popularity of social posts. As a baseline result, the base features achieved SRC and MAE values of 0.65 and 1.38, respectively.

In accordance with Section 3.2, follow [13], we extracted the first type of supplementary features (such as the number of fans, followers, total number of photos posted, etc) for users through data crawling to better represent their characteristics. Upon including these features, the SRC and MAE metrics improved by 0.045 and 0.09, respectively, indicating their effectiveness.

In addition to the statistical characteristics mentioned above, to enhance the semantic information contained in the tags, we employ GloVe [20] to generate tag embeddings. The experiments

demonstrate that the inclusion of tag embeddings resulted in a performance boost of 0.035 for SRC and 0.059 for MAE, respectively. This suggests that tag embeddings can effectively capture the underlying semantic relationships among tags.

Based on these static features, we hypothesize that there exist temporal variations in the overall post popularity at different times. Therefore, we augment our feature set with basic time features such as day of the week and time of day. After adding time series features, the MAE and SR could improve with 0.013 and 0.058. This is consistent with our previous assertion that popular posts exhibit differences across varying time intervals.

The textual and image components of posts contain rich content information that significantly impacts post popularity prediction. Initially, we corrected inconsistencies between the images and text. By introducing similarity features for images and text, we observed improvements in the performance metrics of SRC and MAE by 0.002 and 0.012, respectively. This is in line with our prior assertion that user-generated text may differ from the visual content published alongside it.

Next, we input the text or images of the post separately into the representation learning model to obtain their embedding representations, and then utilize them to obtain the prediction results on the training set and the test set in cross-validation based on the embedding. The prediction results are adopted as stacking features. Despite the strong performance of the aforementioned features, the addition of stacked features resulted in further improvements in SRC and MAE metrics by 0.019 and 0.039, respectively. This highlights the efficacy of the stacking block module in enabling CatBoost to overcome the constraint of being unable to utilize high-dimensional embeddings as direct inputs.

Finally, despite the already impressive accuracy achieved by incorporating stacking features into our model, we observed that the existing features fail to capture user preferences for posting behavior and photo data. To alleviate this issue, as mentioned in Section 3.2, we crawled two more types of data that capture user interactions and use them as supplementary features, named user interaction features. Specifically, the SRC and MAE metrics improved by 0.004 and 0.03, respectively, highlighting the significant contribution of user interaction features to the prediction of social post popularity.

5 CONCLUSION

In this paper, we present a novel approach for social post popularity prediction, named "ECSF". We leverage a range of techniques for feature extraction and construction to enrich the representation of both post and user features, thereby facilitating more accurate predictions of post popularity by the model. To more accurately capture the impact of text and images on post popularity, we introduce an innovative method of constructing stacking features. The results of our experiments highlight the essential contribution of stacking features to predicting post popularity. We conducted massive experiments on the SMP official dataset [26], and achieve the new state-of-art result on the SMP challenge at ACM Multi-media 2023.

¹<http://smp-challenge.com/dataset>

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